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NI-NLM – Lecture 4

LLM inference

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10 Mar 2026

From training to text generation

Recap: training

Model stages:

random neural network

1

"autocomplete on steroids"

base / foundational model

2

assistant

instruction-tuned model

3

helpful assistant

Training stages:

1 **Pre-training** 🌍

Prague is the capital of Czechia (...)

2 **Instruction tuning** 💬

user: What is the capital of Czechia?

assistant: **Prague**

3 **Human preference optimization** 🧑‍⚖️

user: What is the capital of Czechia?

answer #1: Prague.

answer #2: The capital of Czechia is Prague. It is the largest (...)

LLM inference

This lecture: **LLM inference** = we have a trained model and we want to **use it**.

Question

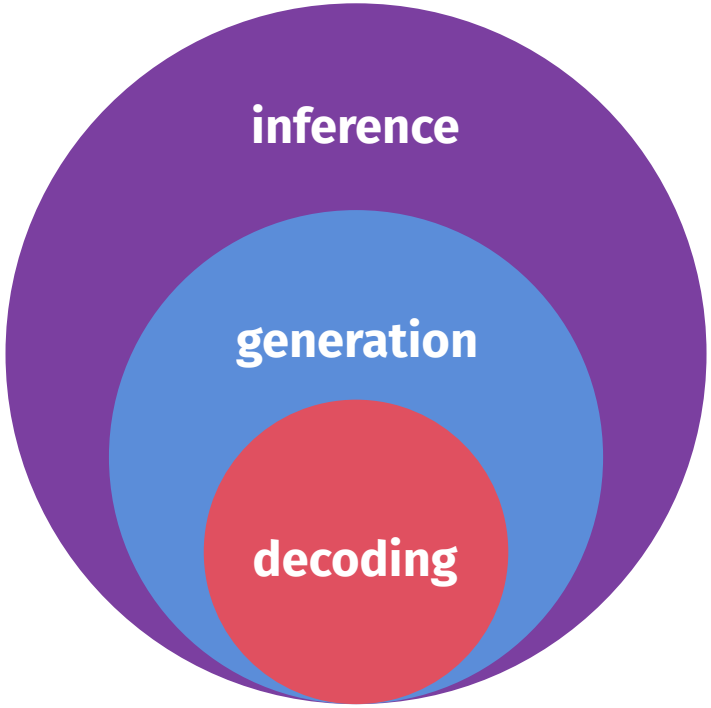
What is the difference between inference, generation, and decoding?

LLM Inference

Inference – The concept of using a trained model for **making predictions** on new data (for classification, sequence tagging, text generation, ...).

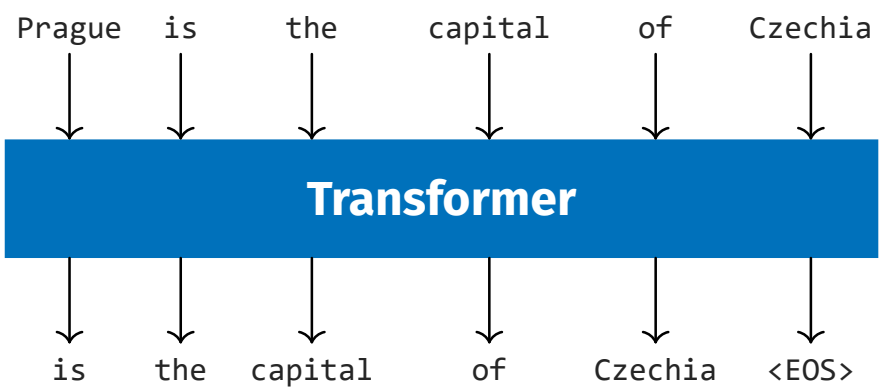
Generation – The process of using a trained model for **producing a sequence of tokens**.

Decoding – The algorithm of **selecting the next token** using the model's internal representation.



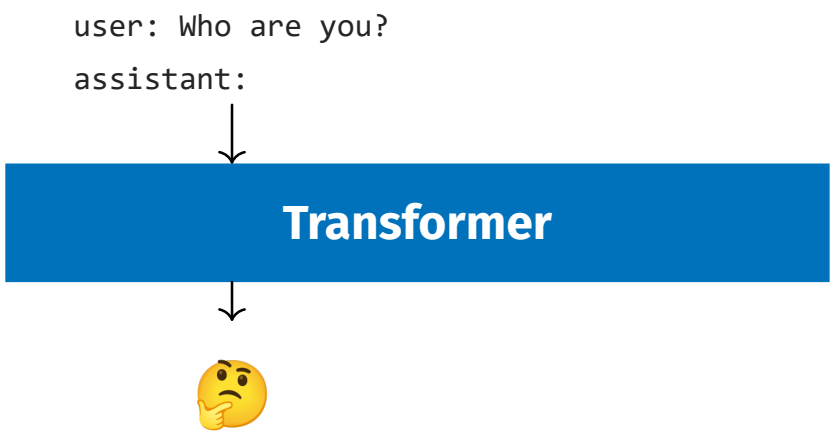
Training vs. inference

Training



Teacher forcing: We know what token should come next, so we use it to train the model.

Inference



Decoding: We need to *select* what token should come next.

Generating text

For each time step t, the decoder outputs **probability distribution** over the tokens given the previous context:

$$P(y_t \mid y_{1:t-1}, X)$$

Question

How to use the distribution to generate text?

Generating text: autoregressive decoding

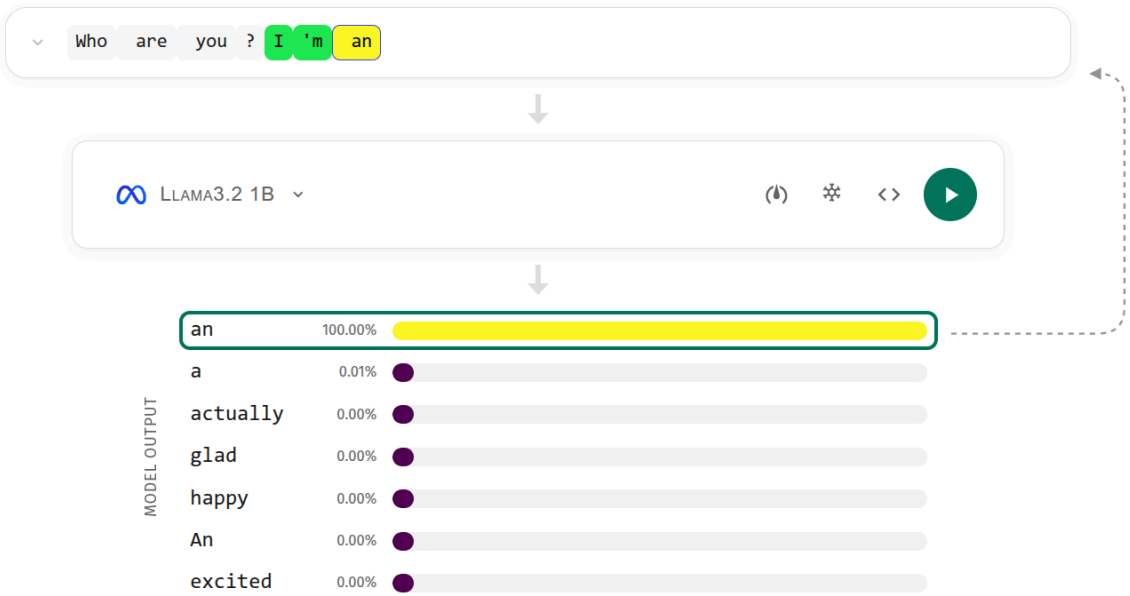
🧑‍🍳 Recipe: How to generate text

1. Start with a sequence of tokens (**prompt**).
2. Feed the sequence into the LLM.
3. Select the next token from the model-generated probability distribution.
4. Append the selected token to the sequence.
5. Repeat from (2).

Note: we are interested only in the **last position's output** – the probability distribution for the next token to generate.

What happens during LLM inference?

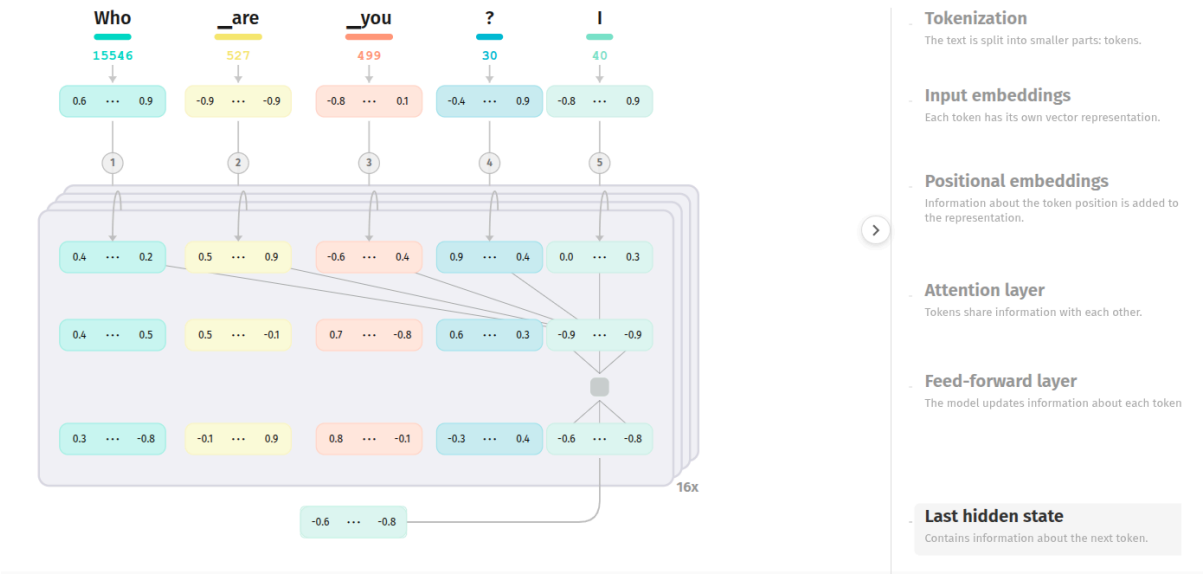
Inference for dummies 🐥



https://animatedllm.github.io/generation-simple

What happens during LLM inference?

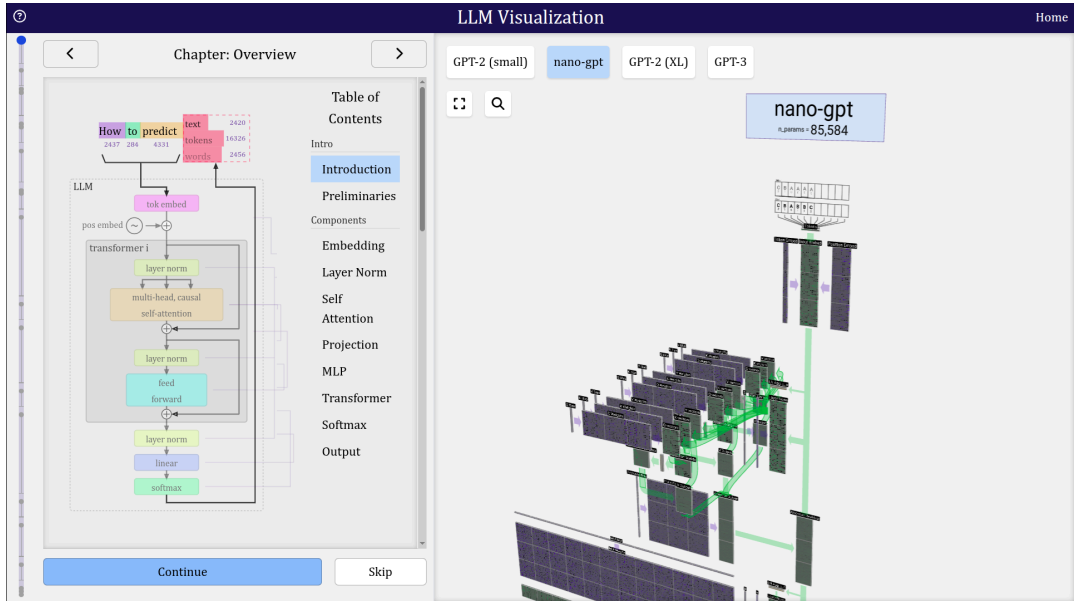
Advanced version 🎓



https://animatedllm.github.io/generation-model

What happens during LLM inference?

Hardcore version 🤘



https://bbycroft.net/llm

Decoding algorithms

Decoding algorithms

For each time step t, the decoder outputs a probability distribution:

$$P(y_t \mid y_{1:t-1}, X)$$

How to use it?

1. **Exact inference**: Find sequence maximizing $P(y_{1:T} \mid X)$.
   - Not possible in practice (why? and is it our goal?).
2. **Approximating the most probable sequence**:
   - Greedy search, beam search.
3. **Adding stochasticity**:
   - Random / top-k / nucleus (Top-p) sampling.

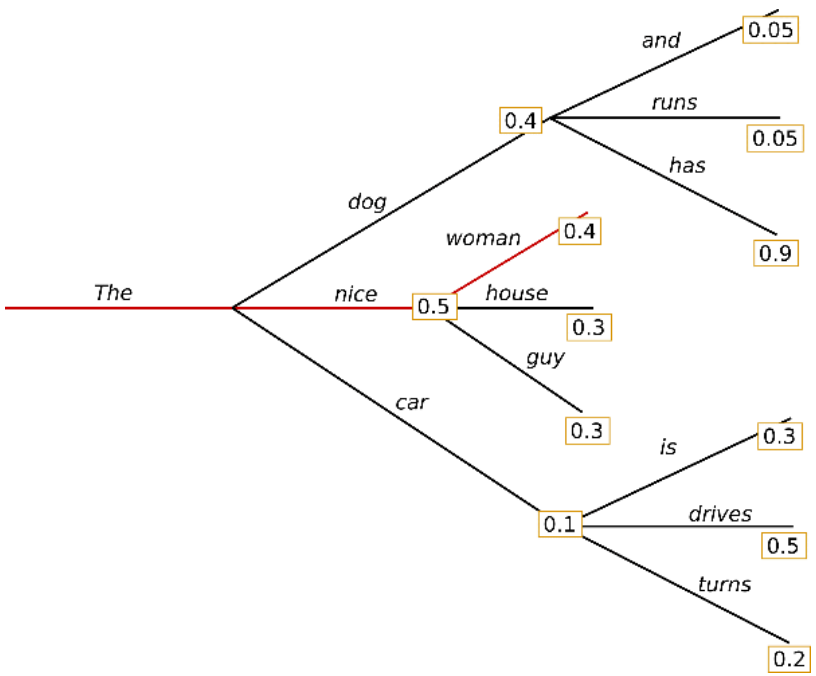
Greedy decoding

source: HF Blog

Selecting the **most probable token in each step** t:

$$y_t = \arg\max_{y_t \in \mathcal{V}} P(y_t \mid y_1, \ldots, y_{t-1})$$

- Very fast, often works satisfactorily (especially with LLMs).
- Non-parametric (no hyperparameters to tune).
- But: may miss high-probability words hidden behind a low-probability word.



Beam search

source: HF Blog

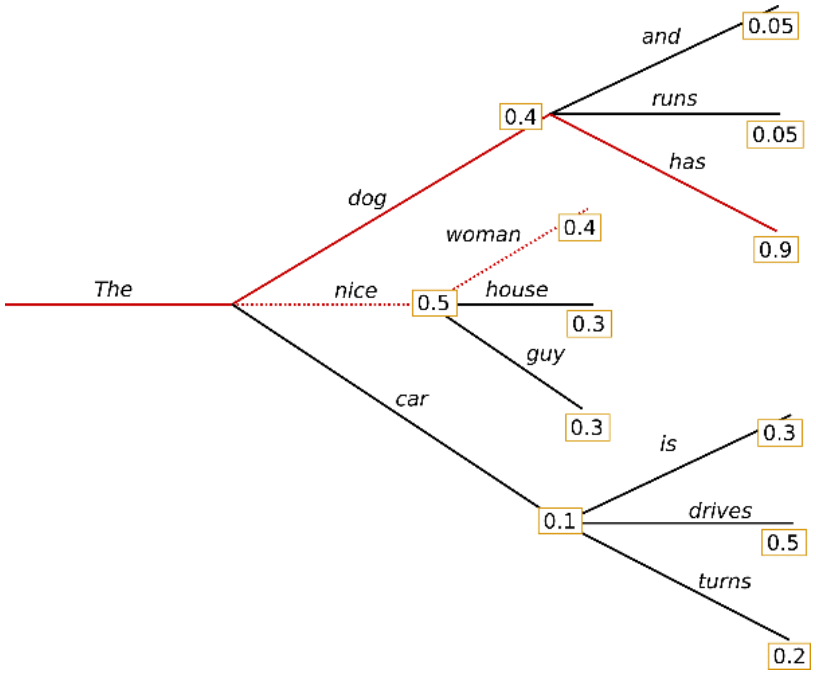
Parameter k: number of sequences (beams).

Each step t:

1. Extend the sequences from step $t-1$ with all possible tokens.
2. Select the k most probable sequences for step $t+1$.

Tuning k:

- $k = 1$ → greedy decoding.
- Larger k → slower, but better approximation.
- $k > 1$ allows re-ranking results.



Exact inference (MAP decoding)

Finding the most probable sequence (= mode of the LM distribution):

$$y^* = \arg\max_{y \in \mathcal{Y}} P(y) = \arg\max_{y \in \mathcal{Y}} \prod_{i=1}^{t} P(y_i \mid y_1, \ldots, y_{t-1})$$

- **Intractable** (exponential search space).
- Can be approximated by greedy decoding or beam search.
- The mode may **not be a good solution**! (e.g. an empty sequence)

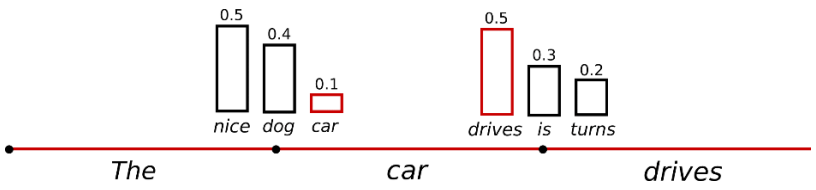
Adding stochasticity

source: HF Blog

In its most basic form, **sampling** means randomly picking the next word w_t according to its conditional probability distribution:

$$w_t \sim P(w \mid w_{1:t-1})$$

- Language generation using sampling is **not deterministic** anymore.
- Human language does not always follow the highest-probability path – we want some surprise!
- But: pure sampling from the full distribution can produce incoherent text.

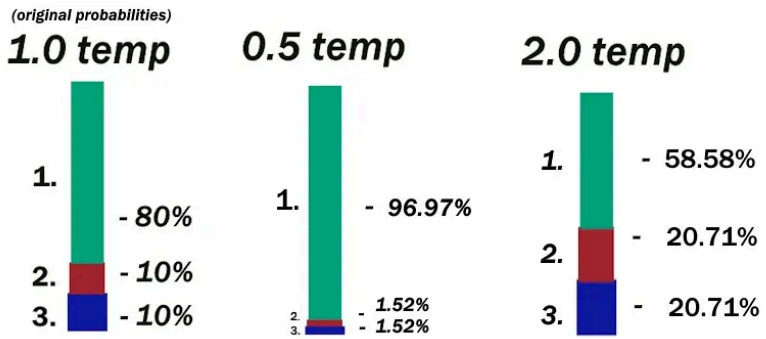


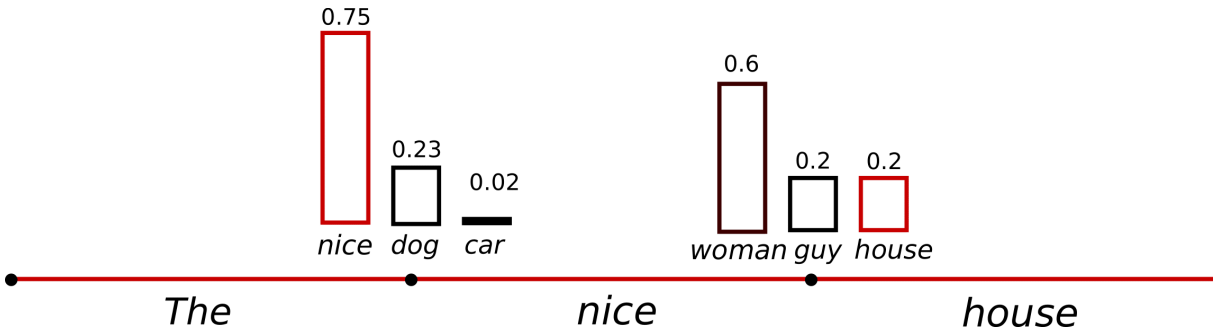
Temperature

The shape of the distribution can be adjusted using the **temperature** T:

$$\mathrm{softmax}(y_i) = \frac{\exp(y_i/T)}{\sum_j \exp(y_j/T)}$$

- $T = 1$: unchanged distribution.
- $T < 1$: **sharper** distribution (more confident).
- $T > 1$: **flatter** distribution (more random).
- $T \to 0$: greedy decoding.



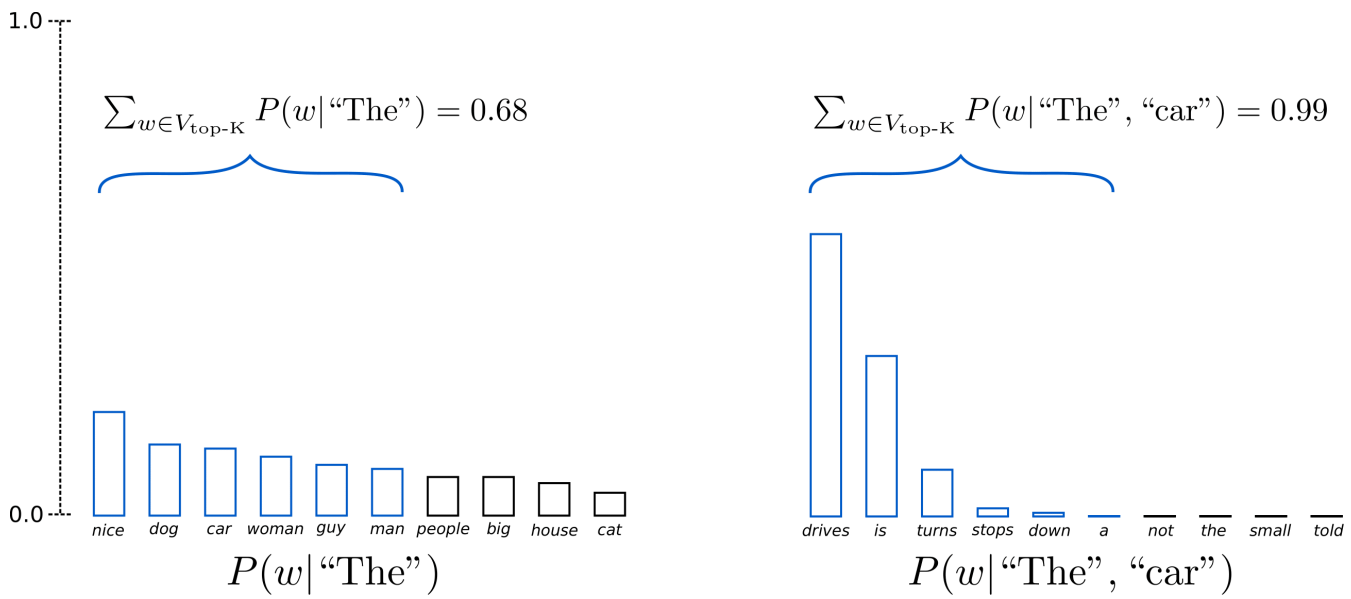

0.75
0.23
0.02
0.6
0.2
0.2
nice
dog
car
woman
guy
house
The
nice
house

Top-k sampling

source: HF Blog

Selecting the token in each step randomly from $k \in \{1, ..., |\mathcal{V}|\}$ most probable tokens.

The truncated distribution is re-weighted using softmax.



Info

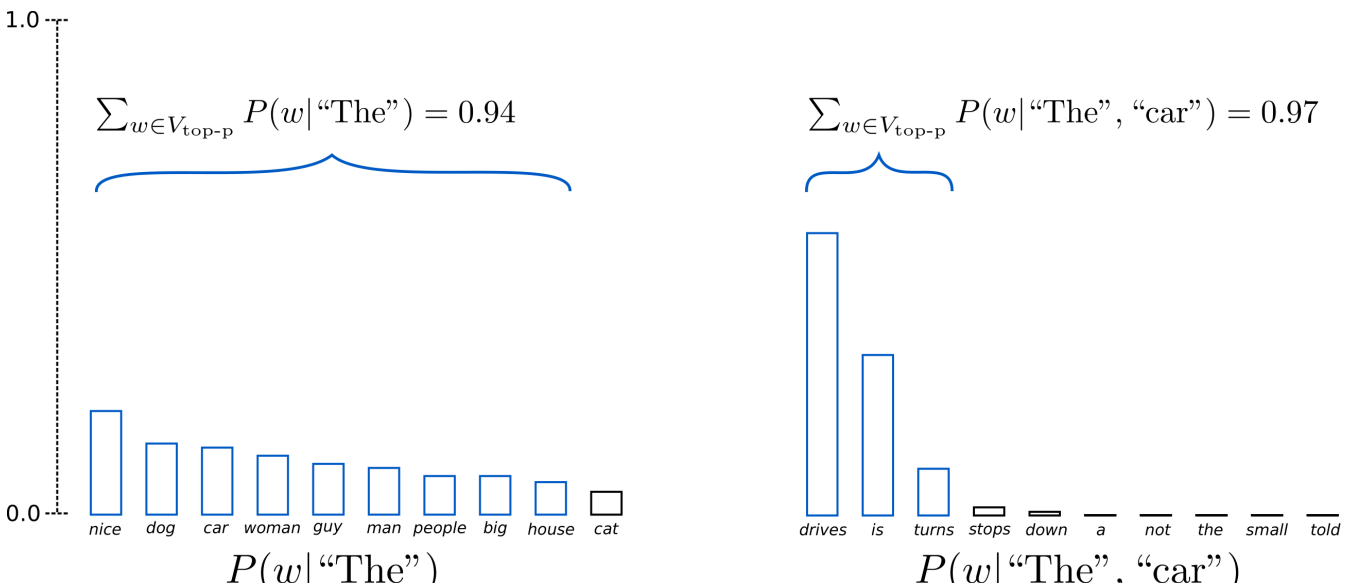
Top-k does not dynamically adapt the number of filtered words – this can be problematic for both sharp and flat distributions.

Top-p (nucleus) sampling

source: HF Blog

Sampling from the **nucleus**: set of the most probable tokens with combined probability summing to $p \in (0, 1]$.

Similar to top-k, but with a **variable** k in each step.



Info

Top-p keeps a wide range of words when the next word is less predictable, and only a few words when it is more predictable.

Is greediness all you need?

r/MachineLearning • 8 mo. ago
zyl1024

[D] What happened to "creative" decoding strategy?

Discussion

For GPT-2 and most models at that time, the naive greedy decoding is extremely prone to generating repetitive and nonsensical outputs very fast, and many techniques, such as top-p sampling, nucleus sampling, repetition penalty, n-gram penalty, etc. are needed. (e.g. https://arxiv.org/pdf/1904.09751)

For recent LLMs, I haven't been using any of these tricks, and instead, any temperature between 0 and 1 seems to work just fine. The only repetitive generation that I've observed seem to be in math reasoning, when the model wants to do some exhaustive search that didn't succeed.

So are all these custom decoding strategies a thing of the past, and we don't need to worry about degenerate content generation anymore?

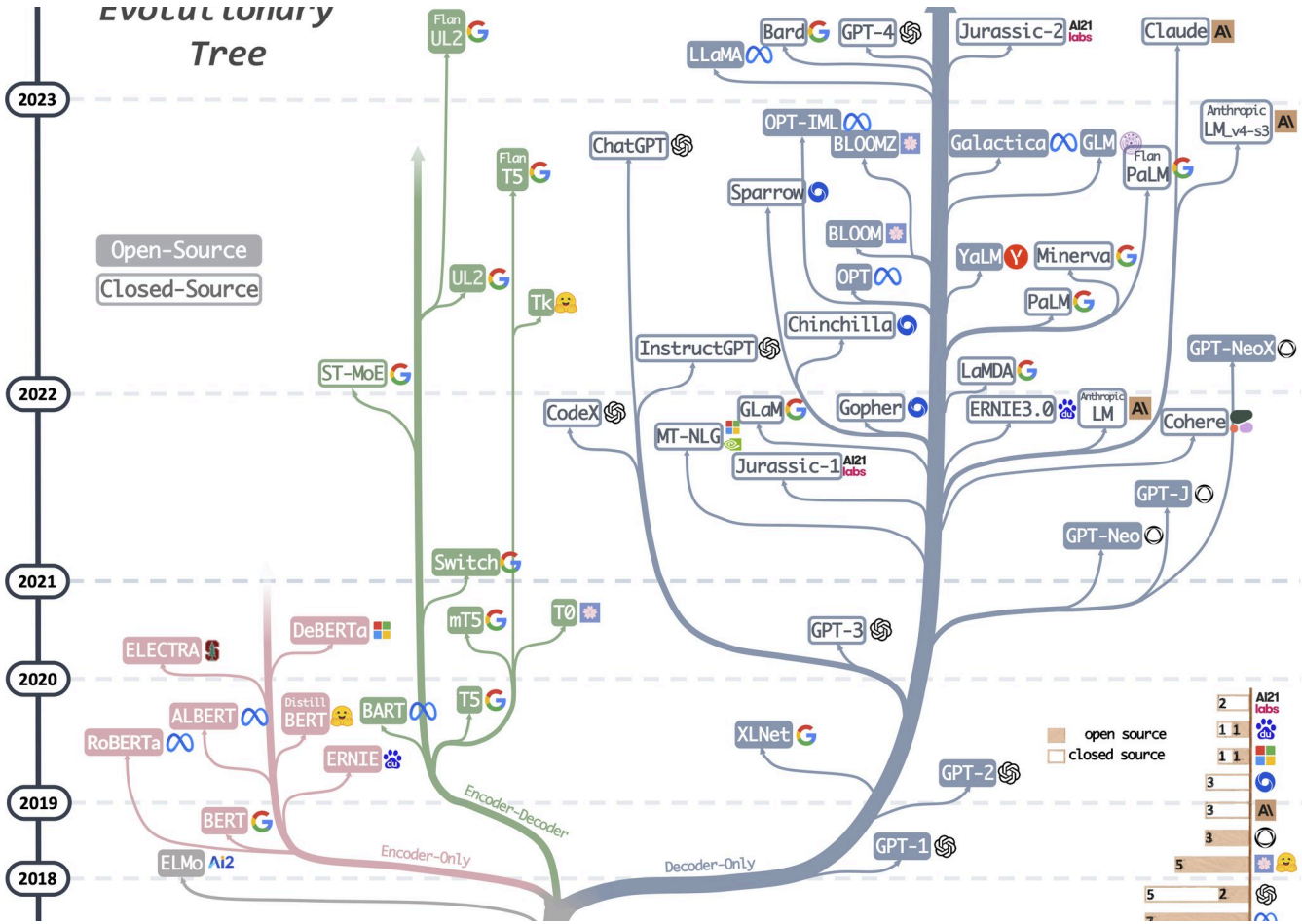
23 12 Share

source: Reddit

Navigating the LLM zoo

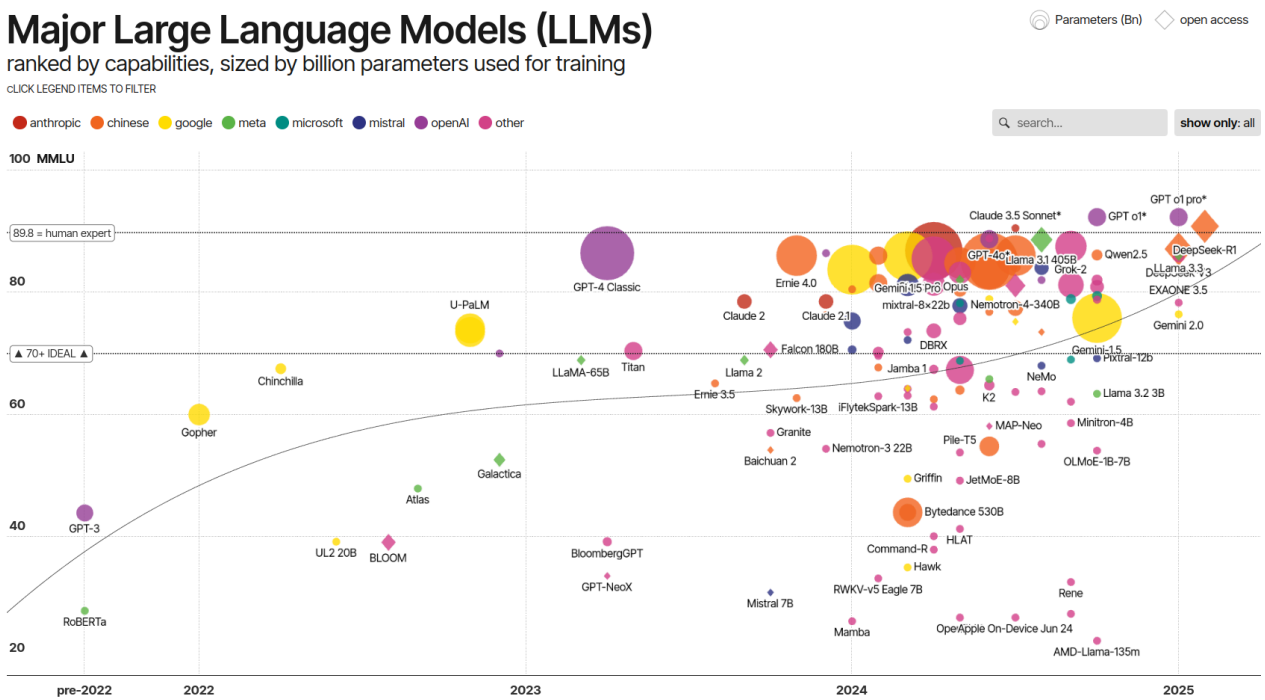
LLM evolutionary tree

source: Yang et al. (2023)



LLM size and capabilities

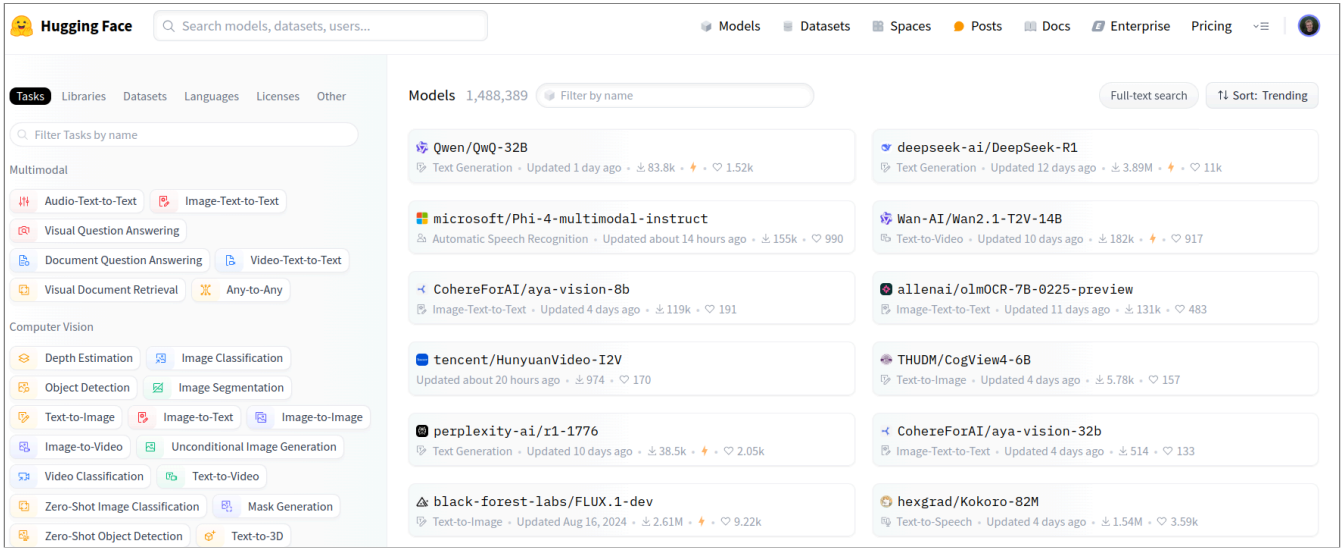
source: informationisbeautiful.net



Model sources and leaderboards: HuggingFace

source: HuggingFace Models

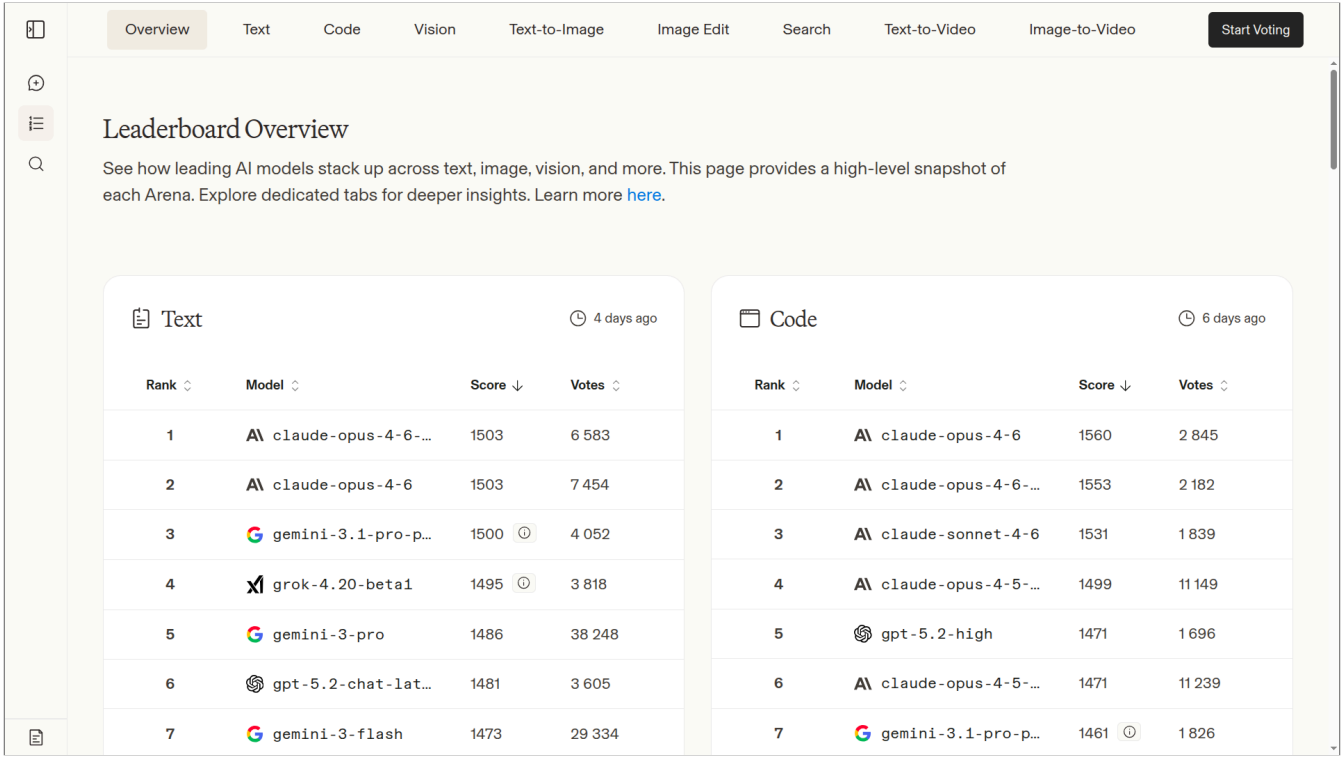
HuggingFace: the largest repository of open LLMs. As of March 2026, it contains 1.5M models (many of these are derivatives).



Model sources and leaderboards: Arena.ai

source: Arena.ai

Arena.ai: Elo rating of LLMs. For a pair of answers from different models, users decide which is better.



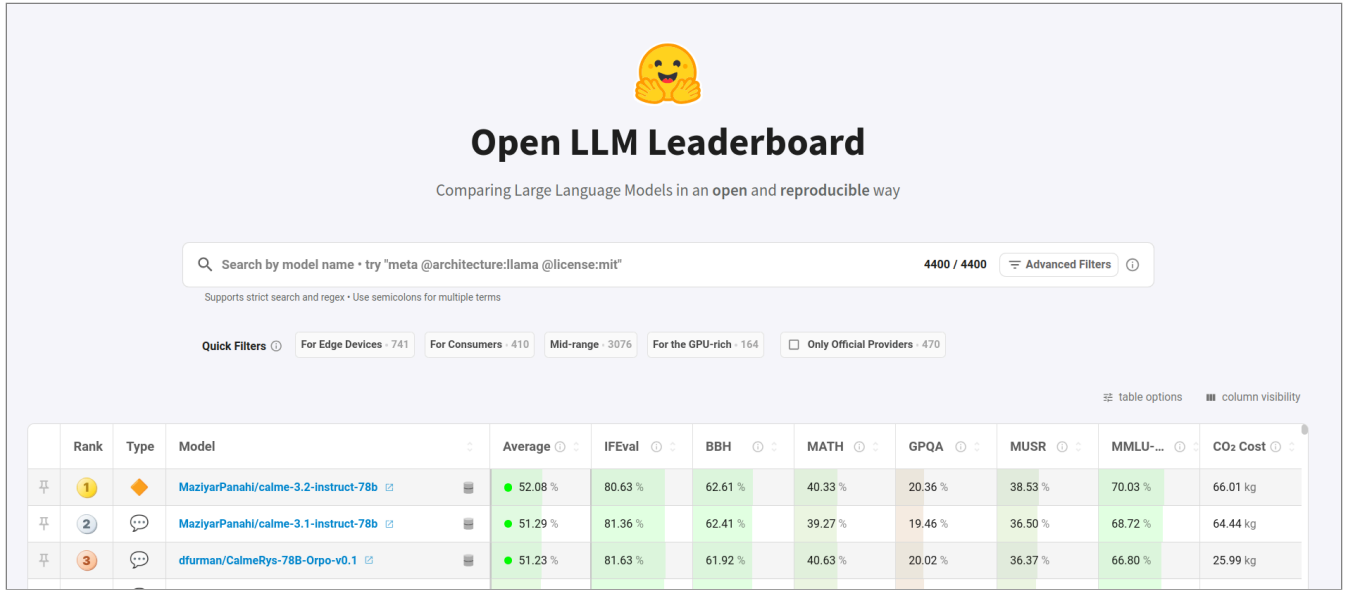
Rank	Model	Score	Votes
1	claude-opus-4-6-...	1503	6 583
2	claude-opus-4-6	1503	7 454
3	gemini-3.1-pro-p...	1500	4 052
4	grok-4.20-beta1	1495	3 818
5	gemini-3-pro	1486	38 248
6	gpt-5.2-chat-lat...	1481	3 605
7	gemini-3-flash	1473	29 334

Rank	Model	Score	Votes
1	claude-opus-4-6	1560	2 845
2	claude-opus-4-6-...	1553	2 182
3	claude-sonnet-4-6	1531	1 839
4	claude-opus-4-5-...	1499	11 149
5	gpt-5.2-high	1471	1 696
6	claude-opus-4-5-...	1471	11 239
7	gemini-3.1-pro-p...	1461	1 826

Model sources and leaderboards: Open LLM Leaderboard

Source: Open LLM Leaderboard

Open LLM Leaderboard: ratings of open LLMs on benchmarks.



Rules of thumb for selecting a model

- Try a **general-purpose model** first.
- You can specify your task using **in-context learning**.
- **RAG** can help you with a custom knowledge base.
- You may want to use a **fine-tuned model**, but think carefully about which data it was finetuned on.
- You probably do **not** want an off-the-shelf **base model** unless you want to fine-tune it (or you are interested in LM on its own).
- Out of the newest models, select the **largest model you can support**.

Running LLMs locally

How to use LLMs

Proprietary APIs

- OpenAI (ChatGPT, GPT-4o)
- Anthropic (Claude)
- Google (Gemini)

✅ Easy to use, no hardware needed

❌ Paid, no control over the model

Open models (local)

- Meta (Llama)
- Mistral
- Qwen, DeepSeek, …

✅ Free, full control

❌ Requires hardware (GPU)

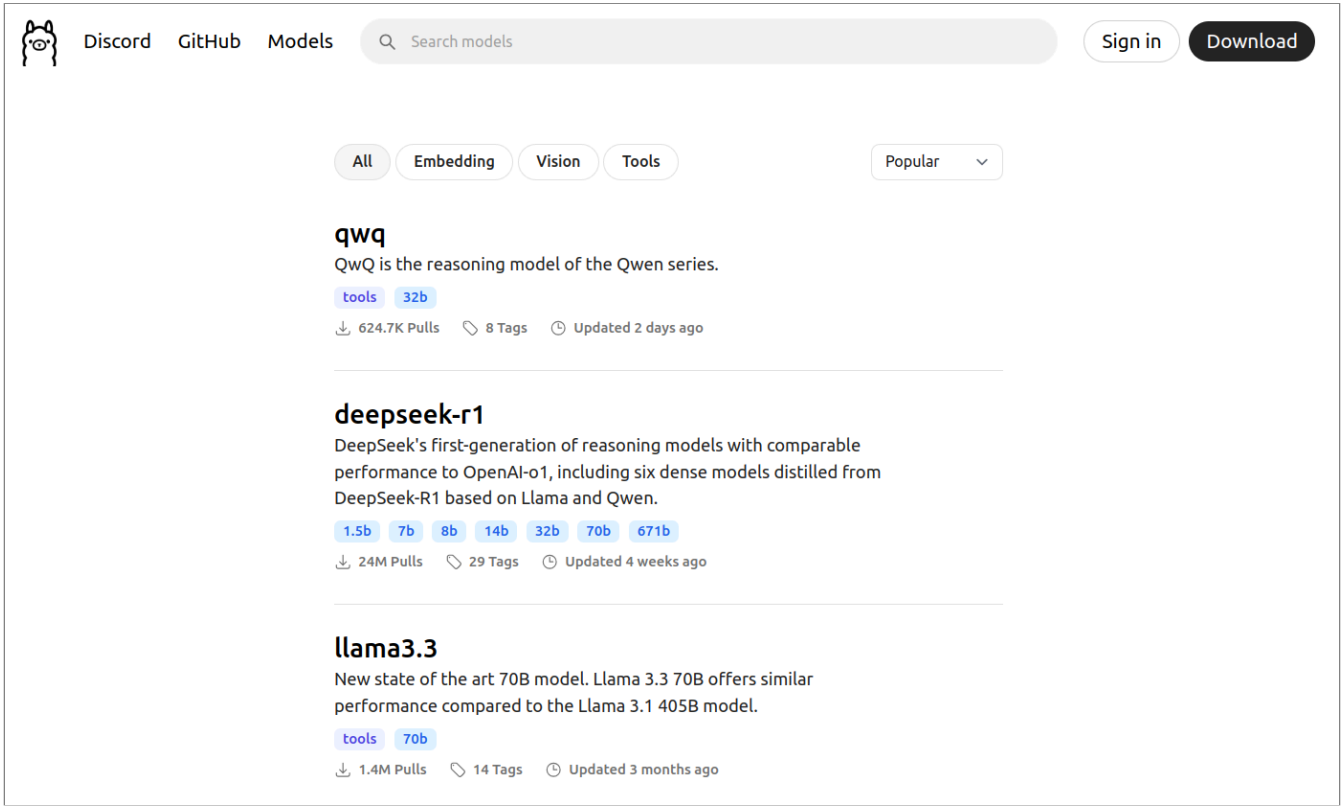
Frameworks for running open LLMs: HuggingFace transformers

HuggingFace transformers: Python library for loading models from the HuggingFace model repository.

Frameworks for running open LLMs: Ollama

source: Ollama

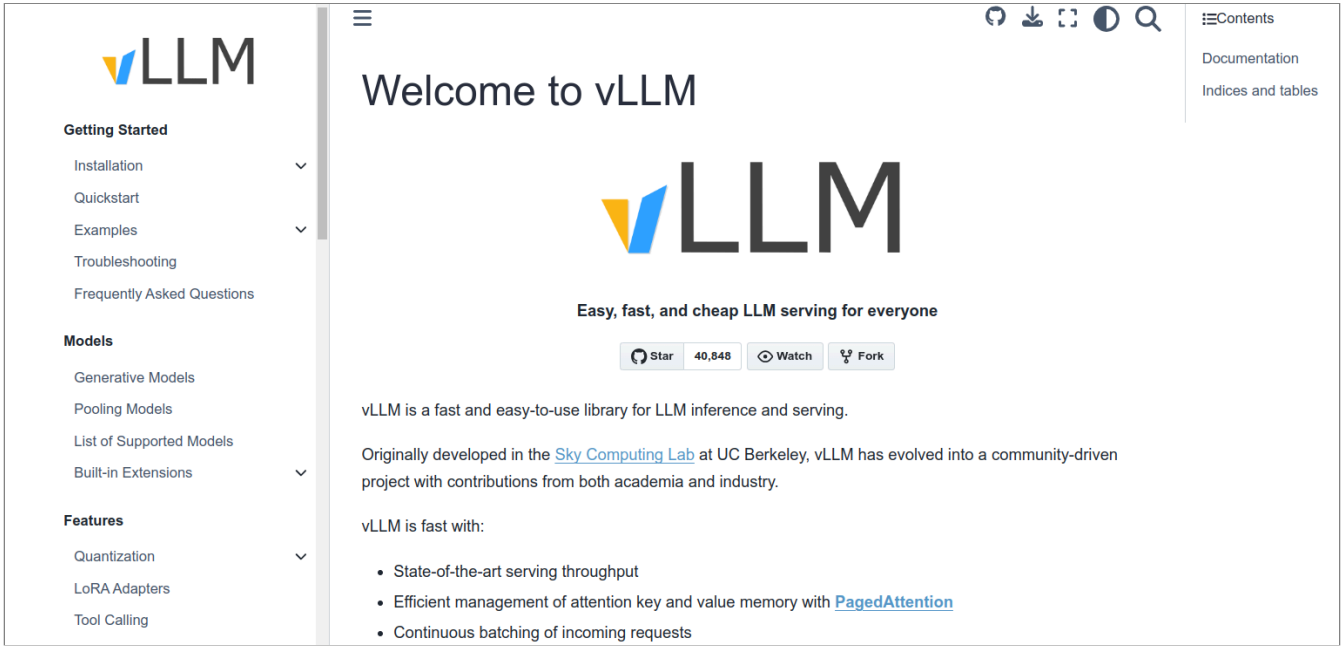
Ollama: running a local server, easy to use, focus on quantized models.



Frameworks for running open LLMs: vLLM

source: vLLM

vLLM: efficient library for serving of LLMs on an enterprise level.



Text generation

Demo time 👩‍💻

👉 HuggingFace LLM tutorial

👉 Decoding strategies tutorial

Links and resources

- HuggingFace models
- Awesome LLM: curated list of resources
- Transformer inference: 3D visualization
- HuggingFace decoding algorithms overview
- HuggingFace text generation strategies
- Common pitfalls when generating text with LLMs
- Visualizing decoding strategies
- Minimum Bayes Risk decoding

Bonus: extra decoding algorithms

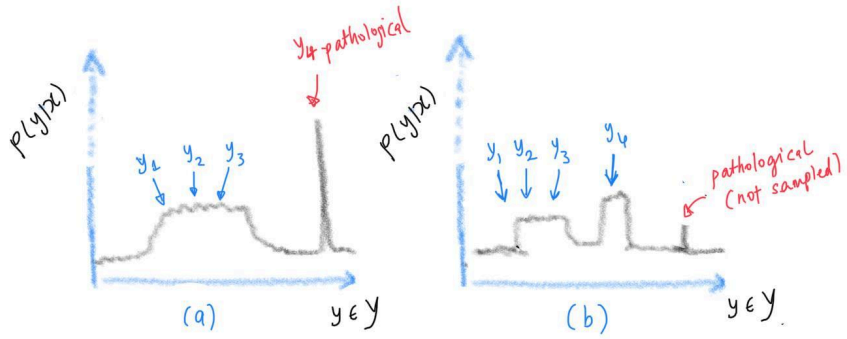
Minimum Bayes Risk (MBR) decoding

source: Minimum Bayes Risk Decoding

Selecting the sequence **most similar to other sequences** = "consensus decoding":

$$y^* = \arg\max_{y_k \in \mathcal{Y}} \sum_{y_\ell \in \mathcal{Y} \setminus \{y_k\}} \mathrm{sim}(y_k, y_\ell)$$

- Useful for minimizing pathological behavior.
- Intractable → we need a sampling algorithm.
- Application in ASR and machine translation.


p(y|x)
y4 pathological
y1 y2 y3
(a)
y ∈ Y
p(y|x)
y1 y2 y3 y4
pathological (not sampled)
(b)
y ∈ Y

Mirostat

source: Basu et al. (2021)

Aims to eliminate **repetition and incoherent text** in stochastic algorithms.

Adapting the k parameter based on the desired text perplexity ("mirum" = surprise, "stat" = control).

Parameters:

- τ – the target perplexity
- η – learning rate

Algorithm 1: Adaptive top-k sampling for perplexity control

Target cross entropy τ, maximum cross entropy $\mu = 2 * \tau$, learning rate η

while *more words are to be generated* **do**

  Compute $\hat{s}$ from (40): $\frac{\sum_{i=1}^{N-1} t_i b_i}{\sum_{i=1}^{N-1} t_i^2}$

  Compute k from (41): $k = \left(\frac{\hat{\epsilon} 2^{\mu}}{1 - N^{-\hat{\epsilon}}}\right)^{\frac{1}{\hat{s}}}$

  Sample the next word X using top-k sampling

  Compute error: $e = \mathfrak{S}(X) - \tau$

  Update μ: $\mu = \mu - \eta * e$

end

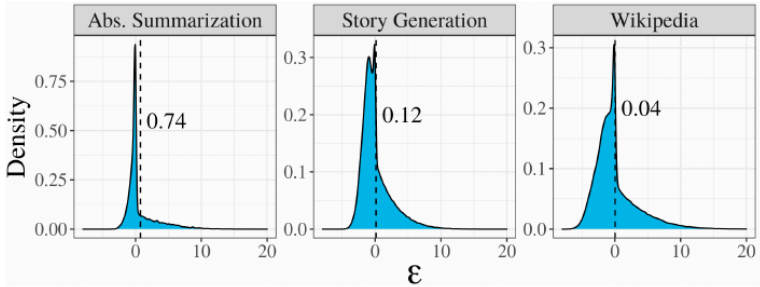
(Locally) typical sampling

source: Meister et al. (2023)

Decodes text so that in each step, its perplexity is **close to the perplexity of the model**.

Similar to Mirostat, but **dynamic**: the perplexity is not pre-specified.

Information theory: **typical messages** are the messages that we would expect from the process.



$p(H) = 0.75$, $p(T) = 0.25$:

- $HHHH$ → most probable sequence
- $HTHH$ → typical sequence

Further reading

- **On decoding strategies for neural text generators** (Wiher et al., 2022) – language generation tasks vs. decoding strategies.
- **If beam search is the answer, what was the question?** (Meister et al., 2020) – why does beam search work so well?
- **Understanding the properties of Minimum Bayes Risk decoding in neural machine translation** (Müller and Sennrich, 2021) – when can MBR be useful?

Summary

Summary

- LLM **inference** = using a trained model for predictions; **generation** = producing a sequence of tokens; **decoding** = selecting the next token.
- **Greedy decoding** is the simplest approach, **beam search** improves it by keeping k hypotheses.
- **Stochastic methods** (top-k, top-p, temperature) add randomness to avoid repetitive and dull outputs.
- There are many LLMs to choose from – use **leaderboards** and **benchmarks** to navigate.
- Open LLMs can be run locally using **HuggingFace transformers**, **Ollama**, or **vLLM**.